A Conceptual Framework for a Visual Reflexivity Engine Dashboard

1. Introduction: The Visual Reflexivity Engine Concept

1.1. Defining Visual Reflexivity for Modular AI Systems

The concept of reflexivity, fundamentally a recursive process of turning back upon oneself or the world, permeates various scientific domains, including cybernetics and systems theory.[1] In the context of control systems, particularly those involving artificial intelligence (AI), reflexivity entails the system's capacity (or the capacity enabled by external observation) to construct models of its own state, behavior, and interactions, potentially including models of other interacting systems, whether natural or artificial.[3] This capability becomes crucial as AI systems increase in complexity and autonomy, demanding mechanisms for self-awareness or, more practically, system-awareness mediated through human observation and interaction.[1]

Applying this to modular AI architectures, such as the polyfunctional chains involving sequential processing stages (e.g., Retrieve, Analyze, Decide, Communicate - RADC), reflexivity implies observing and understanding the internal dynamics and transformations occurring within and between modules. However, the inherent opacity of many AI components, especially deep learning models, often hinders direct introspection.[5] This necessitates external tools and interfaces to facilitate this reflexive process.

This report introduces the concept of **Visual Reflexivity**: the application of interactive data visualization techniques to enable expert users to perceive, interpret, and diagnose the internal states and dynamic processes of complex, modular AI systems. Within this framework, the dashboard acts as more than a passive monitoring interface; it becomes a critical component in a **second-order cybernetic loop**.[3] The technical user (e.g., AI auditor, systems theorist) assumes the role of the observer, utilizing the dashboard to gain insights into phenomena like semantic drift, failure propagation, and value transformation. These visualized insights inform diagnoses and can lead to interventions (e.g., refining prompts, adjusting module parameters, flagging data quality issues), thereby influencing the system's subsequent behavior. This cycle of observation, interpretation, diagnosis, and potential intervention, mediated by the visualization platform, embodies the principle of reflexivity applied to the human-AI system interaction, fostering a deeper understanding and potentially improving the system's alignment and robustness.[1]

1.2. Purpose: Enabling Diagnostic Clarity for Technical Users

The primary purpose of the proposed Visual Reflexivity Engine (VRE) dashboard is to provide **diagnostic clarity** for technical users tasked with overseeing, auditing, or developing complex AI systems. The target audience includes systems theorists, AI auditors, prompt architects, AI safety researchers, and advanced system administrators who require granular insights into the internal workings of these systems, far beyond standard performance metrics like overall accuracy or task completion rates.

Modern AI systems, particularly those constructed from interconnected modules (modular AI [7]) or involving multiple interacting agents [2], often exhibit complex, emergent behaviors and failure modes that are difficult to predict or understand from external observation alone.[19] The internal decision-making processes can be opaque, operating within high-dimensional spaces or complex logic chains.[5] The VRE dashboard aims to pierce this opacity by visualizing key internal dynamics.

The overarching goal is not merely to monitor system status but to facilitate deep diagnostic analysis.[21] By visualizing phenomena such as the subtle degradation of meaning (semantic drift), the occurrence and propagation of specific failure types (e.g., KPI hallucination, logic divergence), the transformation of abstract values (like epistemic certainty or semantic alignment), and the health status of individual modules, the VRE intends to empower users to perform root cause analysis, identify systemic vulnerabilities, and make informed decisions regarding system tuning, maintenance, or redesign.

1.3. Scope: Visualizing Semantic Drift, Failure Modes, and System Health in the RADC-BL Chain

This conceptual framework focuses on visualizing the internal dynamics of a polyfunctional modular AI system, using the RADC-BL (Retrieve, Analyze, Decide, Communicate - Broker, Learner) chain, referenced in prior simulation reports, as a representative architecture. This chain exemplifies a common pattern in complex AI workflows where information is sequentially processed and transformed by specialized modules.

The scope of the VRE dashboard encompasses the visualization of several critical phenomena within such a chain:

1. **Semantic Drift:** Tracking how the meaning or semantic representation of key concepts (e.g., "user engagement," "operational efficiency") changes or degrades as information passes through the RADC-BL modules.[24]
2. **Failure Modes:** Identifying and mapping specific types of failures to the modules or inter-module interfaces where they originate or manifest. This includes failures like KPI hallucination (generating spurious metrics), logic divergence (deviation from expected reasoning paths), data poisoning effects, or adversarial attack impacts.[29]
3. **System Health:** Monitoring the operational status and stability of individual modules using indicators such as module entropy (as a proxy for ambiguity or variance) and confidence mismatch (discrepancy between self-reported confidence and actual performance).[34]
4. **Value Flow and Transformation:** Visualizing the progression and transformation of abstract "value" constructs, such as epistemic value (e.g., certainty, justification [39]) or semantic alignment (fidelity to intended meaning), across the module chain.
5. **Cognitive Interference:** Detecting and visualizing instances of conflict between modules, such as module outputs overwriting each other, generating contradictory information regarding the same concept, or pursuing conflicting goals or assumptions.[15]

The framework presented here is conceptual; implementation details, specific algorithms for calculating metrics (beyond illustrative examples), and the underlying AI system's instrumentation are considered prerequisites rather than outputs of this design phase.

2. Foundational Visualization Principles for Complex AI Dynamics

Designing effective visualizations for the internal dynamics of complex AI systems requires careful consideration of the unique challenges posed by abstract concepts, non-conserved quantities, and multifaceted failure modes. Standard visualization techniques often need adaptation to provide genuine diagnostic clarity.

2.1. Representing Value Flow and Transformation (Epistemic Value, Semantic Alignment)

A core challenge is visualizing the flow and transformation of abstract concepts like "epistemic value" or "semantic

alignment" through the modular chain. Epistemic value can refer to the justification, certainty, or truth-conduciveness of the information held or processed by a module.[39] Semantic alignment refers to the fidelity of the information's meaning relative to an intended concept or ground truth. Unlike physical quantities like energy or material flow, these "values" are not necessarily conserved; they can degrade (e.g., semantic drift [24]), be lost entirely (semantic information loss [63]), transform in meaning, or even be spuriously inflated (e.g., through hallucination).

Standard flow visualization techniques like Sankey diagrams are excellent for showing the magnitude of flow and how it subdivides or merges between different stages or nodes.[74] However, their typical application assumes conservation of the flowed quantity. To adapt them for non-conserved AI "value," enhancements are needed.

Proposed Technique: Enhanced Sankey Diagrams:
Represent the RADC-BL modules as nodes. Links between nodes represent the flow of a specific tracked concept (e.g., "user engagement analysis"). The width of a link can represent the current magnitude of the associated value (e.g., a quantified semantic similarity score to a reference definition, or a measure of epistemic certainty). Crucially, to show transformation or degradation, the color or gradient along the link can be used to encode the change in value occurring across the interface between modules. For instance, a link might transition from green (high initial value) to yellow or red (lower value after processing by the next module), visually representing value decay or semantic drift. Diverging flows from a node can illustrate how value associated with a concept might be interpreted or processed differently by subsequent parallel pathways, if applicable. Tooltips can provide precise numerical values for the value metric and its change.79 Furthermore, if the value metric itself possesses uncertainty (e.g., a probabilistic confidence score), techniques for visualizing uncertainty on edges, such as blurring, dashing, or varying edge texture, can be incorporated.80 This adaptation allows the Sankey metaphor of flow to be applied while explicitly accounting for the non-conservative nature of semantic or epistemic value in AI systems.
Alternative/Complementary Technique: Flow Graphs:
Inspired by tools visualizing information flow within neural networks like transformers 88, a node-link graph can represent the modular system. Modules are nodes, and edges signify information transfer. Edge attributes (thickness, color, texture) can encode the magnitude and change in value, while node attributes (size, color) could represent the internal state of the module or its impact on the value transformation. This approach offers flexibility in layout and potentially allows for representing more complex, non-linear interactions than a standard Sankey.

2.2. Mapping Failure Modes (Semantic Drift, KPI Hallucination, Logic Divergence)

Visualizing failure modes requires associating specific types of failures [29] with their points of origin or manifestation within the modular architecture. A critical aspect is understanding and visualizing **error propagation**.[89] Errors introduced in one module can cascade through the system, causing different types of failures downstream.[19] Cascading failures represent a significant risk in interconnected systems.[111] The visualization must therefore support not just locating failures but tracing their potential paths.

Proposed Technique: Annotated Node-Link Diagrams:

A node-link diagram representing the RADC-BL chain 80 serves as the base. Failures are overlaid using annotations:

- **Icons/Labels:** Distinct icons placed on or near nodes (modules) or links (interfaces) signify specific failure types (e.g., a 'drifting compass' for semantic drift, a 'ghost' for hallucination, diverging arrows for logic divergence).
- **Color-Coding:** Nodes or links can be color-coded based on failure severity (e.g., yellow for warning, red for critical) or the likelihood of failure.
- **Propagation Paths:** When a user selects a detected failure (e.g., hallucination at module L), the visualization should highlight potential upstream causal paths (e.g., A -> D -> C -> B -> L) based on trace data or causal inference models.[132] This directly addresses the need to understand error propagation.[89] Tooltips provide access to detailed error logs and context.[130]

Alternative/Complementary Technique: Heatmaps/Matrix Views:
A heatmap 79 can provide an aggregated view of failure patterns. A matrix with modules on one axis and failure types (or time) on the other can visualize failure hotspots. Cell color intensity would represent the frequency or severity of a specific failure type within a particular module over a selected time window, helping identify chronic issues.

The crucial element is moving beyond simply marking where a failure was *detected* to supporting the diagnosis of where it *originated*. This requires linking the structural view (the graph) with temporal and causal information derived from system traces.

2.3. Visualizing Cognitive Interference (Module Overwrites, Conflicts)

Cognitive interference arises when modules produce conflicting outputs, assumptions, or follow divergent reasoning paths regarding the same concept or goal.[15] This can manifest as direct overwrites, where one module's output negates another's, or as more subtle conflicts stemming from differing interpretations or underlying assumption mismatches.[15] Such conflicts are particularly relevant in multi-agent systems where agents might have explicitly conflicting goals or beliefs.[15]

Proposed Technique: Specialized Link/Node Visualizations:
Within the node-link diagram framework:

- **Conflict Links:** Draw specialized links (e.g., thick dashed red lines, lines with 'spark' animations) directly between nodes representing modules that have generated conflicting outputs or signals related to the same concept.
- **Node/Interface Indicators:** Use specific icons (e.g., 'opposing arrows', 'warning triangle') overlaid on the conflicting nodes or on the links representing the data flow where the conflict manifests.
- **Interaction:** Hovering or clicking on a conflict indicator should reveal details about the conflicting data points, the specific outputs from each module involved, and potentially links to the modules' configurations or input data that might explain the conflict.

Alternative/Complementary Technique: Parallel Coordinates / Comparison Views:
When multiple modules output comparable metrics (e.g., confidence scores, risk assessments) for the same item, parallel coordinates plots 77 can effectively highlight divergences. Each axis represents a module, and lines connecting the values for a specific data instance across modules make disagreements visually apparent.

It is important to recognize that observed conflicts (like differing outputs) are often symptoms of deeper issues,

such as conflicting baseline assumptions programmed into the modules or divergent interpretations of ambiguous upstream data.[52] While visualizing the surface conflict is necessary, a truly diagnostic tool should ideally provide pathways to investigate these underlying causes, potentially by linking the conflict visualization to views showing module configurations or input data provenance. Furthermore, visualizing the frequency or persistence of conflicts over time (e.g., by varying the thickness or color intensity of conflict links based on historical data) can help distinguish between isolated incidents and systemic incompatibilities.

2.4. Incorporating Uncertainty Visualization

Many metrics within AI systems are inherently uncertain or probabilistic. Confidence scores, semantic similarity measures, failure likelihood predictions, and even the data flowing between modules can carry uncertainty. Failing to represent this uncertainty can lead to misinterpretations and potentially flawed decision-making by the user.[80] Effective uncertainty visualization techniques are therefore essential.[80]

Applicable Techniques:

- **Uncertainty on Edges:** For visualizing uncertainty associated with the flow or relationship represented by an edge (e.g., uncertainty in a value drift metric, probabilistic failure propagation), techniques include:
  - **Visual Variables:** Varying edge blurriness (more blur = more uncertainty), using dashed lines with gap size proportional to uncertainty, modifying edge shape (e.g., waves with amplitude indicating uncertainty), or using textures.[80]
  - **Encoding Distribution:** Using glyphs along the edge, varying edge thickness or color saturation/lightness to represent confidence intervals or variance.
- **Uncertainty on Nodes:** For uncertainty related to a module's state or the location of a failure:
  - **Glyphs:** Small, dedicated symbols placed within or near the node to encode uncertainty distribution or level.[80]
  - **Visual Variables:** Modifying node color saturation/lightness, using dashed or blurred node borders, varying node size slightly.
  - **Embedded Plots:** For nodes representing aggregated data, embedding small statistical plots like box plots [80] or histograms within the node.
- **Uncertainty in Scalar Values/Time Series:** For metrics like confidence scores, entropy, or failure likelihood shown in panels or tooltips:
  - **Explicit Ranges:** Error bars showing standard deviation or confidence intervals.[82]
  - **Distribution Displays:** Confidence bands around line charts [82], violin plots [83], density plots [83], or histograms showing the probability distribution.
- **General Approaches:** Using transparency/opacity to indicate certainty (more transparent = more uncertain) [82], providing explicit probability values as text annotations, or linking summary views to detailed distribution visualizations in separate panels.[80]

The choice of technique should depend on the nature of the uncertainty being visualized (e.g., a probability value, an interval, a full distribution [84]) and the analytical task the user needs to perform (e.g., identifying high-uncertainty areas, comparing uncertainty levels between modules, understanding the shape of an uncertain distribution). The goal is not just to indicate the presence of uncertainty but to make the uncertainty itself interpretable and actionable.

Table 1: Visualization Techniques for AI System Dynamics

Phenomenon to Visualize	Primary Technique	Secondary/Complementary Techniques	Key Supporting Evidence	Justification & Considerations
Value Flow & Transformation	Enhanced Sankey Diagram [74]	Flow Graphs [88], Node-Link Diagrams with attribute mapping	75	Sankey excels at flow quantity & stages but needs enhancement (e.g., color/gradient on links) for non-conserved value transformation. Flow graphs offer structural alternatives.
Failure Mode Mapping & Propagation	Annotated Node-Link Diagram [80]	Heatmaps (Failure Hotspots) [79], Time-series plots (Failure Evolution)	89	Node-link shows structure; annotations (icons, color) map failures. Heatmaps reveal patterns. Crucial to visualize propagation paths [89], not just detection locations.
Cognitive Interference (Conflicts/Overwrites)	Specialized Link/Node Visualizations (Node-Link) [15]	Parallel Coordinates [77], Comparison Views	17	Direct visual cues (special links, node icons) signal conflict. Deeper visualization may be needed for underlying causes (e.g., assumption mismatch [52]). Parallel coordinates useful for comparing conflicting outputs.
Uncertainty	Edge/Node Visual Variables [80], Glyphs [80], Statistical Plots [80]	Error Bars [82], Confidence Bands [82], Transparency [82], Linked Distribution Views [80]	80	Multiple techniques exist. Choice depends on uncertainty type (value vs. location) and task. Goal is interpretable representation of uncertainty itself [86], not just indicating its presence.

3. Component Design: Graph - Value Drift | Failure Type | Module Overwrites

This central component of the VRE dashboard aims to provide an integrated view of the AI system's structure, the flow and transformation of value through it, and the occurrence of failures and conflicts.

3.1. Visualizing the RADC-BL Chain Structure

The foundation of this component is a **Node-Link Diagram**.[80] Each module in the RADC-BL sequence (Retrieve,

Analyze, Decide, Communicate, Broker, Learner) is represented as a distinct node. Directed links connect these nodes, representing the primary pathway of data or information flow between consecutive modules.

The **layout** of this graph is crucial for conveying the system's logic. Given the sequential nature of the RADC-BL chain, a directed graph layout, such as a layered or hierarchical approach (e.g., Sugiyama-style), is recommended to clearly emphasize the processing flow from input (at R) to output (potentially from C or L).[17] This makes it easy for users to follow the progression of information. While force-directed layouts [126] can reveal community structures, they might obscure the intended sequence and are likely less suitable unless exploring complex, non-linear interactions between modules is a primary goal. If modules contain internal sub-components, a hierarchical layout allowing node expansion could be beneficial.[131] This clear structural representation serves as the canvas upon which dynamic operational data is overlaid.

3.2. Representing Value Drift

Value drift refers to the undesired alteration of semantic meaning or the degradation of epistemic value [39] as information traverses the modular chain. Semantic drift, specifically, is the change in meaning [24], which can be quantified using metrics that measure semantic similarity between representations of a concept at different stages.[139]

Quantification Strategy:

1. Identify key concepts whose semantic integrity is critical (e.g., "user intent," "risk assessment," "efficiency metric").
2. At each interface between modules (e.g., R output -> A input), extract the representation (e.g., text, embedding) of the tracked concept.
3. Calculate a semantic similarity score between the concept's representation at the current interface and its representation at the previous interface (to measure incremental drift) or against a canonical, ground-truth definition (to measure absolute drift). Suitable metrics include BERTScore [139] or models fine-tuned for Semantic Textual Similarity (STS) like STSScore.[142]
4. Define a Value Drift Metric, for example, as 1 - Similarity Score or the change in similarity score across the interface. Higher values indicate greater drift.

Visualization on the Node-Link Graph:

- **Link Attributes:** The calculated drift metric for a specific concept flowing across an interface can be mapped to visual attributes of the corresponding link.
  - *Color:* Use a continuous color scale (e.g., green for low drift, yellow for moderate, red for high drift) to encode the drift magnitude.
  - *Thickness:* Vary link thickness (e.g., thicker lines for higher drift).
  - *Gradient:* Apply a color gradient along the length of the link to visually suggest the drift occurring *across* the interface boundary.
- **Node Attributes:** Node appearance (e.g., size, border color intensity) could represent the cumulative drift experienced by concepts processed *within* that module, or perhaps an aggregated measure of the module's tendency to introduce drift based on historical data.
- **Interaction:** Users should be able to select one or more concepts to track, dynamically updating the link visualizations to show drift specific to those concepts. Tooltips on links must display the specific drift score, the underlying similarity metric used, and the concept being tracked.

Visualizing the *magnitude* of drift is essential. A small, acceptable drift might be colored green, while a significant semantic shift triggering downstream issues would be bright red, immediately drawing the user's attention to problematic interfaces or modules.

3.3. Indicating Failure Types and Locations

Failures such as semantic drift (which can be both a cause and effect related to value drift), KPI hallucination [32], logic divergence [30], or other system faults [33] need to be clearly mapped onto the system structure. Critically, the visualization must support tracing **error propagation** [89], distinguishing between where a failure is *detected* and where it might have *originated*.

Visualization on the Node-Link Graph:

- **Icons/Labels:** Standardized, easily recognizable icons should be placed directly on the node (module) or link (interface) associated with a detected failure event. For example, a 'ghost' icon for hallucination on the Learner module, or a 'drifting compass' icon on the link between Analyze and Decide if significant semantic drift is detected there. Concise text labels can supplement icons if space allows.
- **Color-Coding:** Use color on node borders or backgrounds (or link colors) to indicate the presence and severity of active failures (e.g., yellow outline for a warning-level failure, solid red fill for a critical failure).
- **Error Propagation Path Highlighting:** This is crucial for diagnosis. When a user selects a failure indicator (e.g., the hallucination icon on module L), the system should leverage trace data or causal models [132] to highlight the likely upstream path of the problematic information flow.[89] For instance, the links and nodes connecting A -> D -> C -> B -> L might be highlighted, suggesting the hallucination originated from drift introduced in module A.
- **Interaction:** Clicking a failure icon or a highlighted element must open a detailed view (e.g., in the Details-on-Demand panel) showing specific error logs, timestamps, associated input/output data snippets, and any available diagnostic information. Filtering options should allow users to view only specific failure types or failures above a certain severity threshold across the entire graph.

This approach enables users to quickly identify failing components but, more importantly, provides visual support for tracing failures back to their potential root causes, which is essential for effective remediation.

3.4. Signaling Module Overwrites and Conflicts

Conflicts arise when modules produce contradictory outputs, operate under conflicting assumptions, or when one module's actions negate or override another's intended contribution.[15] These issues are vital to visualize for understanding internal system friction.

Visualization on the Node-Link Graph:

- **Conflict Links:** When a conflict is detected between the outputs or states of two specific modules (say, A and D) regarding the same data instance or concept, draw a distinct type of link directly between the nodes A and D. This link could be styled differently (e.g., red, dashed, jagged, perhaps with an animated 'spark' effect) to clearly distinguish it from normal data flow links.
- **Node Indicators:** Place a specific 'conflict' icon (e.g., opposing arrows, a warning triangle with an exclamation mark) directly onto the nodes involved in the conflict or overwrite event.
- **Interface Indicators:** Alternatively, or additionally, place conflict indicators on the outgoing links from the conflicting modules, specifically on the pathways where the conflicting information is propagated.
- **Interaction:** Hovering over or clicking on a conflict link or node indicator should trigger a display (e.g., in a

tooltip or the Details panel) showing the specific data points or concepts involved in the conflict, the conflicting values or outputs generated by each module, and timestamps.

It is also valuable to visualize the persistence of conflicts. A conflict link's thickness or color intensity could dynamically represent the frequency or duration of conflicts logged between those two modules over the selected time window. This helps differentiate transient, perhaps noise-induced, disagreements from persistent, systemic incompatibilities that might point to fundamental issues in module design, coordination logic, or underlying assumptions.[52]

4. Component Design: Color-Coded Interface - Module Health

This component provides a summarized view of the health and stability of individual modules within the RADC-BL chain, using color-coding for quick assessment. It focuses on metrics that go beyond simple error counts, aiming to capture aspects of module ambiguity, self-awareness, and susceptibility to specific critical failures.

4.1. Visualizing Module Entropy

Module entropy serves as a proxy for the level of 'disorder,' ambiguity, instability, or unpredictability in a module's behavior or output.[34] High entropy can signal potential problems, even if the module isn't overtly failing according to predefined error conditions.

Calculating Entropy Proxies: Direct calculation of thermodynamic entropy is generally not feasible. Instead, practical proxies must be used:

- **Output Ambiguity:** For modules producing probabilistic outputs (e.g., classifiers), the Shannon entropy of the output probability distribution can be calculated. Higher entropy indicates less certainty in the prediction.[34] H(p)=−Σipilog2pi.
- **Performance Variance:** Track key performance metrics for the module (e.g., processing latency, resource consumption, accuracy on validation subsets) over time or across different input types. High variance or standard deviation in these metrics can indicate instability.[146]
- **Deviation from Expected Behavior:** If a baseline model of 'normal' behavior exists (e.g., a learned distribution of typical outputs for given inputs), the deviation of current outputs from this baseline (e.g., measured by KL divergence or other distance metrics) can serve as an entropy proxy.
- **Semantic Output Variance:** For modules generating embeddings or textual outputs, calculate the variance in the semantic space (e.g., average pairwise cosine distance between embeddings) for outputs generated from semantically similar inputs. High variance suggests inconsistent semantic processing.

Visualization:

- **Heatmap:** A compact way to display entropy levels across all modules simultaneously. Each cell represents a module (R, A, D, C, B, L), and the cell's color intensity maps directly to the calculated entropy proxy value (e.g., light = low entropy/stable, dark = high entropy/unstable).[79] This allows for rapid identification of potentially problematic modules.
- **Node Color/Indicator:** In the main graph view (Section 3), the color fill or border of each module node can be mapped to its current entropy level, providing contextual health information alongside the flow and failure data.
- **Time-Series Plot:** For selected modules, display the entropy proxy value over time as a line chart [148] in the Details-on-Demand panel. This helps identify trends, sudden spikes, or periodic instability.

Monitoring module entropy is valuable because high entropy in an upstream module can be a leading indicator of downstream issues. A module producing ambiguous or highly variable outputs injects uncertainty into the processing chain, which can accumulate [99] and eventually manifest as more severe failures like semantic drift or logical divergence in subsequent modules.

4.2. Visualizing Confidence Mismatch

This metric focuses on the calibration of a module's self-assessment capabilities. It highlights discrepancies between a module's reported confidence in its output and the actual, externally assessed quality (performance, correctness, or semantic fidelity) of that output.[45] This assessment might come from comparison to ground truth, evaluation by a dedicated meta-agent or monitoring agent [5], or human auditor feedback.

Quantification:
A simple Mismatch Score can be calculated as:
Mismatch=∣Reported Confidence−Assessed Performance/Fidelity∣
Both confidence and performance/fidelity should be normalized to a comparable scale (e.g., 0 to 1). A more nuanced approach could track the direction of mismatch (over-confidence vs. under-confidence).
Visualization:

- **Color Gradient:** Apply a diverging color gradient to the module node or a dedicated health indicator. For example:
  - Blue: Under-confident (Reported Confidence << Assessed Performance)
  - White/Grey: Well-calibrated (Reported Confidence ≈ Assessed Performance)
  - Red: Over-confident (Reported Confidence >> Assessed Performance) The intensity of the color indicates the magnitude of the mismatch.
- **Glyph:** Design a composite glyph that visually encodes both reported confidence (e.g., size or fill level of one part) and assessed performance (e.g., size or fill level of another part), making the discrepancy immediately visible.
- **Scatter Plot:** In the Details-on-Demand panel, plot Reported Confidence versus Assessed Performance for individual outputs processed by the module over a time window. Points falling far from the y=x diagonal represent significant mismatches.

Identifying confidence mismatch is critical because it reveals how trustworthy a module's own certainty estimates are. Consistent over-confidence is a major risk factor, especially in sequential systems. An upstream module might pass faulty information with high confidence, leading downstream modules to trust and propagate the error.[89] Under-confidence, while potentially less catastrophic, might indicate overly conservative behavior or missed opportunities for effective action. This visualization directly flags modules whose self-assessment reliability is questionable.

4.3. Visualizing KPI Hallucination Likelihood

Key Performance Indicator (KPI) hallucination is a specific failure mode where a module, particularly one involved in reporting, analysis summarization, or learning (like the Broker or Learner modules in the RADC-BL chain), generates spurious, inaccurate, or entirely fabricated KPIs or metrics.[32] This can be triggered by corrupted or ambiguous inputs resulting from upstream semantic drift or processing errors.

Estimating Likelihood:
Since direct detection of hallucination might require external validation, the VRE dashboard can visualize an estimated likelihood of hallucination based on contributing factors:

- **Upstream Semantic Drift:** Higher cumulative semantic drift detected in the inputs to the module.[24]
- **Input Ambiguity:** Higher entropy or uncertainty measured in the module's inputs.[34]
- **Module History:** Historical frequency of confirmed hallucinations from this module under similar input conditions or operational states.
- **Task Complexity:** Reliance on complex reasoning, abstraction, or multi-step inference, which are known to be more prone to LLM failures.[99]
- **Model State:** Internal model indicators associated with known failure modes like objective neglect.[32]

A heuristic score combining these factors can be calculated to represent the current likelihood of KPI hallucination.

Visualization:

- **Color-Coded Risk Level:** Assign a categorical risk level (e.g., Low, Medium, High) based on the estimated likelihood score. Represent this using distinct colors (e.g., Green, Yellow, Red) applied to the module node in the graph view or in the Module Health Panel.
- **Risk Gauge/Meter:** A simple visual gauge or meter associated with the module, showing the current estimated likelihood percentage or level.
- **Alert Icons:** Display a prominent warning icon (e.g., a flashing red light, a specific hallucination symbol) next to the module when the estimated likelihood exceeds a predefined critical threshold.

Visualizing the *likelihood* provides a proactive warning mechanism. It allows users to scrutinize the outputs of modules deemed high-risk for hallucination more carefully, even before a hallucination event is definitively confirmed by external checks. This is particularly important for modules like the Broker or Learner whose outputs might directly influence system strategy or user understanding.

5. Integrated Dashboard Concept: The Visual Reflexivity Engine

The Visual Reflexivity Engine (VRE) dashboard integrates the specialized visualizations for value flow, failures, conflicts, and module health into a coherent interface designed for diagnostic analysis by technical experts. The design prioritizes clarity, interactivity, and the ability to explore relationships between different aspects of the system's behavior, thereby facilitating the reflexive process of understanding and potentially guiding the AI system.

5.1. Proposed Layout and Flow

A multi-panel layout is proposed to balance overview and detail, a common practice in effective dashboard design [21]:

- **Central Pane: System Dynamics Graph:** This area prominently displays the primary Node-Link Diagram detailed in Section 3. It serves as the main navigational and contextual anchor, showing the RADC-BL chain structure overlaid with visualizations of value drift (e.g., link color/thickness), failure occurrences (e.g., icons, node highlighting), and module conflicts (e.g., special links, node indicators). This provides the essential structural and dynamic overview of the system.
- **Left Pane: Module Health Panel:** This panel provides a summarized view of the health indicators for each module (R, A, D, C, B, L), as detailed in Section 4. It could use small multiples (miniature indicators for each module) or a compact heatmap [79] to display Module Entropy, Confidence Mismatch, and KPI Hallucination

Likelihood using color-coding. Selecting a module in this panel could filter or highlight the corresponding node and its connections in the central graph.

- **Right Pane: Details-on-Demand Panel:** This panel remains initially empty or shows global summaries. When a user selects an element (node, link, icon) in the central graph or the health panel, this area populates with detailed information relevant to the selection. This could include specific metric values, time-series plots of selected indicators [148], error logs, snippets of input/output data, module configuration details, or provenance traces.
- **Top Bar: Global Controls & Summary:** This bar houses global controls such as selecting the time range for the displayed data, choosing specific concepts to track for value drift analysis, applying global filters (e.g., show only critical failures), and displaying high-level system summary KPIs.

User Interaction Flow: A typical diagnostic workflow might proceed as follows [21]:

1. **Overview:** User observes the central graph and the module health panel to get a general sense of the system state.
2. **Identify Anomaly:** User spots an area of interest, such as a link with high value drift (bright red), a node flagged with a critical failure icon, a module showing high entropy in the health panel, or a conflict link between two modules.
3. **Select & Investigate:** User clicks on the element of interest. This highlights the element and potentially related elements (e.g., propagation path) in the central graph and populates the Details-on-Demand panel with specific data, logs, or time-series plots related to the selection.
4. **Explore Context:** User examines the detailed information, potentially using tooltips for quick info or analyzing trends in the time-series plots.
5. **Filter & Refine:** User applies filters (e.g., time range, failure type) via the top bar or interactions within the panels to narrow down the analysis and isolate potential causes or correlations.
6. **Iterate:** User selects other related elements or adjusts filters based on findings, continuing the investigation until a diagnosis is formed.

5.2. Interaction Design

Rich interactivity is key to enabling effective diagnostic exploration:

- **Tooltips:** On-hover display of concise information for nodes (module name, summary health status), links (flow direction, current value/drift score), icons (failure type, conflict details), and points on charts (exact value, timestamp).[79]
- **Filtering:** Dynamic filtering of the displayed data based on user selections. Filters could apply to:
  - Time range (global control).
  - Concept being tracked (for value drift visualization).
  - Failure type or severity level.
  - Module health status (e.g., show only modules with high entropy).
  - Specific modules selected in the health panel.[21]
- **Selection & Highlighting:** Clicking a node, link, or indicator should:
  - Visually highlight the selected element.
  - Potentially highlight related elements (e.g., upstream/downstream path for propagation analysis, other nodes involved in a conflict).[130]
  - Populate the Details-on-Demand panel with relevant detailed information.[167]
- **Brushing & Linking:** Coordinated interaction between views. For example, selecting (brushing) a time range

on a time-series plot in the Details panel should filter the central graph view to show the system state during that specific period, and vice-versa.[80] Selecting a module in the Health Panel highlights it in the graph.

- **Zoom & Pan:** Standard interactive navigation for the potentially complex central graph view to examine specific areas in detail.[131]
- **Drill-Down:** If the underlying data supports it, allow users to "drill down" into a module node. Clicking the node could expand it to show internal sub-components or switch the Details panel to show finer-grained internal metrics or processing steps.[23]

5.3. Supporting Diagnostic Analysis for Technical Users

The VRE dashboard's design must explicitly support diagnostic tasks, moving beyond simple status monitoring towards facilitating root cause analysis and system understanding.[132]

- **Visual Analytics Integration:** The design leverages principles of visual analytics by tightly integrating multiple views that present different facets of the data: the structural/flow view (central graph), attribute summaries (health panel), and temporal/detailed views (details panel with logs and time-series).[17] The coordination between these views through linking and brushing is essential for enabling users to connect different pieces of evidence.
- **Inspiration from Related Tools:** Concepts from visual analytics systems designed for complex systems can be adapted. For instance, AgentLens [169] demonstrates techniques for visualizing agent trajectories, hierarchical behavior summaries, and causal traces in multi-agent systems, which could inform how interactions and sequences are presented. Similarly, tools like VISIT [88], which visualize semantic information flow in transformers using interactive flow graphs, offer paradigms for representing the progression of meaning.
- **Facilitating Reflexivity:** The dashboard acts as the interface for the reflexive loop.[1] Its design must make it easy for users to correlate observed phenomena (e.g., a failure flagged in module C) with potential contributing factors visualized elsewhere (e.g., high value drift on the link entering C, high entropy in module A upstream, specific patterns in the input data shown in the details panel). This ability to connect observations across different views and metrics is fundamental to forming diagnostic hypotheses and understanding the system's behavior reflexively.

An effective diagnostic dashboard for complex AI systems requires more than just displaying data; it necessitates a carefully designed integration of multiple, coordinated views and rich interaction capabilities.[132] This allows expert users to navigate the complexity, connect disparate pieces of information, and build a comprehensive understanding necessary for diagnosis and informed action.

6. Data Input Requirements

The conceptual framework for the Visual Reflexivity Engine dashboard relies on the availability of specific data streams from the underlying polyfunctional modular AI system (e.g., the RADC-BL chain) and potentially external assessment mechanisms. The richness, granularity, and quality of these data inputs directly determine the feasibility and diagnostic power of the proposed visualizations. Defining these requirements is crucial for implementing both the AI system instrumentation and the dashboard itself.

The following data inputs are considered necessary or highly desirable for the VRE dashboard to function as intended:

1. **Module Input/Output Data:** The actual content (or representative samples/summaries) of the data packets

passed between consecutive modules (R->A, A->D, D->C, C->B, B->L). This provides context for understanding module operations and diagnosing failures.

2. **Concept Representations:** For key concepts being tracked (e.g., "user engagement"), the specific representation (e.g., embedding vector, structured data, text snippet) of that concept as it exits one module and enters the next. Needed for calculating value drift.
3. **Semantic Similarity Scores:** Pre-calculated or calculable scores (e.g., using BERTScore [139] or STSScore [142]) measuring the similarity between concept representations at consecutive interfaces or against a canonical definition. This directly feeds the value drift visualization.
4. **Module Confidence Scores:** Each module's self-reported confidence level associated with its primary output(s) for a given input. Required for the confidence mismatch visualization.
5. **Assessed Performance/Fidelity Scores:** An external measure of the actual correctness, quality, or semantic fidelity of each module's output. This could originate from a meta-agent/auditor agent [5], comparison with ground truth data, results from offline evaluation pipelines, or potentially structured human feedback. Needed for confidence mismatch calculation.
6. **Module Performance Metrics:** Standard operational metrics like processing latency, throughput, CPU/memory usage per module. Can be used as a proxy for module stability/entropy and displayed in the details panel.
7. **Logged Error/Failure Types:** Timestamped logs generated by the AI system or monitoring agents whenever a specific failure condition is detected (e.g., semantic drift threshold exceeded, hallucination detected by meta-agent, logic divergence identified, hardware fault [33]). Logs should include the module/interface involved, timestamp, severity, and contextual information.[29] Essential for failure mapping.
8. **Conflict/Overwrite Logs:** Timestamped logs indicating instances where conflicting outputs were generated by different modules for the same concept/input, or where one module's output directly overwrote another's.[15] Needed for conflict visualization.
9. **Trace Data (Provenance):** Crucially, data that allows linking a specific output from a module back to the specific input(s) it received, and tracing this lineage back through the entire chain.[199] This is fundamental for visualizing and diagnosing error propagation.[89] Implementing robust provenance tracking requires careful design within the AI system's architecture.
10. **Module Configuration/Prompts:** Access to the current configuration parameters, prompts, or rule sets being used by each module. Provides vital context for interpreting behavior, especially unexpected behavior or drift potentially caused by problematic prompts.[99]
11. **Entropy Proxy Data:** The raw data needed to calculate the chosen proxy metric(s) for module entropy. This might include historical performance metrics, output probability distributions from classifiers, or semantic embeddings of recent outputs.[34]

The feasibility and diagnostic depth of the VRE dashboard are fundamentally constrained by the ability to collect and access these data streams. In particular, robust traceability/provenance information [199] is paramount for moving beyond simple status monitoring to effective root cause analysis of propagated errors and semantic drift.

Table 2: Data Input Requirements for Visual Reflexivity Engine

Data Input	Description	Dashboard Component(s) Enabled	Importance for Diagnosis	Potential Sources

Module Input/Output Data	Content/samples passed between modules.	Details Panel (Context), Value Drift (Concept Extraction), Failure Analysis (Context)	Essential for understanding context of operations and failures.	AI System Instrumentation/Logging
Concept Representations	Embeddings/symbolic forms of tracked concepts at interfaces.	Value Drift Calculation	Needed to quantify semantic similarity/drift.	AI System Instrumentation/Logging
Semantic Similarity Scores	Similarity between concept representations (e.g., using [139]).	Value Drift Visualization (Link Color/Thickness)	Directly quantifies semantic stability/drift magnitude.	VRE Backend (calculated) or External Metric Service
Module Confidence Scores	Module's self-reported confidence in its output.	Confidence Mismatch Visualization, Details Panel	Needed to assess module self-awareness and calibration.	AI System Instrumentation/Logging
Assessed Performance/Fidelity	External measure of output correctness/fidelity.	Confidence Mismatch Visualization, Details Panel	Ground truth for evaluating module reliability and confidence calibration.	Meta-Agent [5], Human Audit Interface, Offline Evaluation, Ground Truth Data
Module Performance Metrics	Latency, throughput, resource usage, etc.	Module Health (Entropy Proxy - Variance), Details Panel	Can indicate module stress or instability (proxy for entropy).	AI System Monitoring Infrastructure
Logged Error/Failure Types	Timestamped logs of specific failures (drift, hallucination, etc.).[29]	Failure Mapping (Icons, Color), Details Panel	Direct evidence of failures, essential for tracking and root cause analysis.	AI System Error Logging, Monitoring Agents [160]
Conflict/Overwrite Logs	Timestamped logs of detected module conflicts/overwrites.[15]	Conflict/Overwrite Visualization (Special Links/Icons), Details Panel	Highlights points of internal friction or contradictory processing.	AI System Instrumentation/Logging, Comparison Logic
Trace Data (Provenance)	Links outputs to inputs across the RADC-BL chain.[199]	Error Propagation Path Visualization, Value Flow Tracking	Crucial for understanding causality and diagnosing propagated errors/drift.[89]	AI System Instrumentation/Logging (requires careful design)
Module Configuration/Prompts	Current settings/prompts for each module.	Details Panel (Context), Failure Analysis (Potential Cause)	Essential context for interpreting module behavior; potential source of drift/failure.[99]	AI System Configuration Management
Entropy Proxy Data	Data for calculating entropy proxies (e.g., output distributions,	Module Entropy Visualization	Needed to quantify module ambiguity/instability.	AI System Instrumentation/Logging, Performance

	metric history).[34]			Monitoring History

7. Conclusion and Future Directions

7.1. Summary of the Conceptual Framework

This report has outlined a conceptual framework for a **Visual Reflexivity Engine (VRE)** dashboard, designed to provide diagnostic clarity into the internal dynamics of polyfunctional modular AI systems, such as the RADC-BL chain. Grounded in the principles of reflexivity [1], the VRE dashboard serves not merely as a monitoring tool, but as an interactive interface enabling technical users to observe, interpret, diagnose, and potentially influence the complex behaviors within these systems.

The framework prioritizes the visualization of phenomena critical for understanding system reliability and alignment, including:

- **Value Flow and Transformation:** Using enhanced Sankey diagrams or flow graphs to track abstract values like epistemic certainty or semantic alignment, explicitly visualizing degradation (drift) or transformation.
- **Failure Modes and Propagation:** Employing annotated node-link diagrams to map specific failures (semantic drift, KPI hallucination, logic divergence) to modules/interfaces and, crucially, to visualize potential error propagation paths.
- **Cognitive Interference:** Utilizing specialized link and node visualizations to highlight conflicts and overwrites between modules.
- **Module Health:** Assessing individual module stability and reliability through color-coded indicators for entropy proxies, confidence mismatch, and KPI hallucination likelihood.
- **Uncertainty:** Incorporating appropriate visual techniques to represent the inherent uncertainty in AI metrics and data.

These elements are integrated into a multi-view dashboard concept featuring a central system dynamics graph, a module health panel, and a details-on-demand panel, interconnected through interactive features like filtering, selection, highlighting, and brushing/linking. The design emphasizes support for diagnostic workflows, enabling users to move from high-level overviews to detailed investigation and root cause analysis. The framework also explicitly defines the necessary data inputs, highlighting the importance of comprehensive system instrumentation, particularly for provenance tracking, to enable the desired level of diagnostic insight.

7.2. Potential Extensions and Research Areas

The proposed VRE framework provides a foundation, but several avenues exist for future extension and research:

- **Predictive Analytics:** Integrate predictive models to forecast potential failures or high-risk states based on current trends in visualized metrics (e.g., rising entropy, accelerating semantic drift).[132]
- **Interactive "What-If" Analysis:** Develop capabilities allowing users to simulate the potential impact of interventions (e.g., changing a module's prompt, adjusting a parameter) directly within the dashboard interface to explore remediation strategies.
- **Advanced Causal Inference:** Incorporate more sophisticated causal discovery algorithms to automatically infer and visualize more robust causal relationships underlying observed failures and propagation patterns [132], moving beyond simple trace highlighting.
- **Automated Remediation Suggestions:** Based on diagnoses facilitated by the dashboard, integrate AI components that suggest potential remediation actions (e.g., specific prompt adjustments, module retraining recommendations, data filtering rules).

- **Generalization to Complex Architectures:** Extend the visualization concepts to handle more complex AI architectures beyond linear chains, such as systems with parallel branches, feedback loops, dynamic agent team formation, or hierarchical structures.[17]
- **Metrics Research:** Further research is needed to develop more robust, interpretable, and computationally efficient metrics for quantifying semantic drift, epistemic value/uncertainty in complex AI outputs, and meaningful proxies for module entropy.
- **User Studies and Evaluation:** Rigorous user studies involving the target audience (systems theorists, AI auditors, etc.) are essential to validate the effectiveness and usability of the proposed visualization techniques and interaction designs for real-world diagnostic tasks.[80] Evaluating how well the dashboard supports the intended reflexive loop would be particularly valuable.

By addressing these areas, the Visual Reflexivity Engine concept can evolve into a powerful toolset for enhancing the transparency, trustworthiness, and effective management of increasingly complex and autonomous AI systems.

Works cited

1. Reflexivity and the History of Psychology - Oxford Research Encyclopedias, accessed on April 21, 2025, https://oxfordre.com/psychology/display/10.1093/acrefore/9780190236557.001.0001/acrefore-9780190236557-e-469?p=emailAmW.uW0/xBjVg&d=/10.1093/acrefore/9780190236557.001.0001/acrefore-9780190236557-e-469
2. Cybernetics - Wikipedia, accessed on April 21, 2025, https://en.wikipedia.org/wiki/Cybernetics
3. Reflexivity and Artificial Intelligence in Control (Subjectness-oriented Approach), accessed on April 21, 2025, https://www.researchgate.net/publication/356178480_Reflexivity_and_Artificial_Intelligence_in_Control_Subjectness-oriented_Approach
4. www.reflexion.ru, accessed on April 21, 2025, https://www.reflexion.ru/Library/Lepskiy2021a.pdf
5. Data Governance for AI Agents: What You Need to Know | Alation, accessed on April 21, 2025, https://www.alation.com/blog/data-governance-for-ai-agents-what-you-need-to-know/
6. Second-order cybernetics - Wikipedia, accessed on April 21, 2025, https://en.wikipedia.org/wiki/Second-order_cybernetics
7. www.ijfmr.com, accessed on April 21, 2025, https://www.ijfmr.com/papers/2025/2/42035.pdf
8. Creating Scalable AGI: the Open General Intelligence Framework - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2411.15832v2
9. Generative AI Architecture: Layers and Models - SoluLab, accessed on April 21, 2025, https://www.solulab.com/generative-ai-architecture/
10. Creating Scalable AGI: the Open General Intelligence Framework - arXiv, accessed on April 21, 2025, https://arxiv.org/pdf/2411.15832?
11. What Is Modular AI Architecture? - Magai, accessed on April 21, 2025, https://magai.co/what-is-modular-ai-architecture/
12. Agent Architectures and Scalability: Building Robust AI Systems - SmythOS,

accessed on April 21, 2025, https://smythos.com/ai-integrations/tool-usage/agent-architectures-and-scalability/
13. Modular and Customizable AI Models - Machine Learning Design ..., accessed on April 21, 2025, https://mlopszone.wyztechsolutions.com/index.php/2025/01/14/modular-and-customizable-ai-models/?amp=1
14. Building AI Agents with LangChain and AutoGPT: A Practical Guide, accessed on April 21, 2025, https://www.amazon.co.uk/Building-Agents-LangChain-AutoGPT-Autonomous/dp/B0DWLHFJTJ
15. Multi-agent systems | The Alan Turing Institute, accessed on April 21, 2025, https://www.turing.ac.uk/research/interest-groups/multi-agent-systems
16. Integrating Multi-Agent Systems and Reinforcement Learning - SmythOS, accessed on April 21, 2025, https://smythos.com/ai-agents/multi-agent-systems/multi-agent-systems-and-reinforcement-learning/
17. What is a Multiagent System? - IBM, accessed on April 21, 2025, https://www.ibm.com/think/topics/multiagent-system
18. Multi-agent system - Wikipedia, accessed on April 21, 2025, https://en.wikipedia.org/wiki/Multi-agent_system
19. Perceptions of Agentic AI in Organizations: Implications for Responsible AI and ROI - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2504.11564v1
20. AgentDynEx: Nudging the Mechanics and Dynamics of Multi-Agent Simulations - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2504.09662v1
21. 30 Proven Dashboard Design Principles for Better Data Display - Aufait UX, accessed on April 21, 2025, https://www.aufaitux.com/blog/dashboard-design-principles/
22. 16 Best Dashboard Design Examples: Ways to Visualize Complex Data - Eleken, accessed on April 21, 2025, https://www.eleken.co/blog-posts/dashboard-design-examples-that-catch-the-eye
23. Effective Dashboard Design Principles for 2025 | UXPin, accessed on April 21, 2025, https://www.uxpin.com/studio/blog/dashboard-design-principles/
24. A General Scalable Approach for Knowledge Selection based on ..., accessed on April 21, 2025, https://www.sciengine.com/doi/10.3724/2096-7004.di.2025.0006
25. Survey in Characterization of Semantic Change - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2402.19088v2
26. Preprint: July 19,2024 - arXiv, accessed on April 21, 2025, https://arxiv.org/pdf/2402.19088
27. Temporal Dynamics in Short Text Classification: Enhancing Semantic Understanding Through Time-Aware Model - MDPI, accessed on April 21, 2025, https://www.mdpi.com/2078-2489/16/3/214
28. Generative AI Decision-Making Attributes in Complex Health ..., accessed on April 21, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC11871968/

29. Failure Prediction for Autonomous Systems - mediaTUM, accessed on April 21, 2025, https://mediatum.ub.tum.de/doc/1651981/1651981.pdf
30. Meta-Coherence Indexing: How Theories Phase-Lock Across AI, Search, and Scientific Discourse - PhilArchive, accessed on April 21, 2025, https://philarchive.org/archive/BOSMIH
31. Failure Modes When Productionizing AI Systems - Robust Intelligence, accessed on April 21, 2025, https://www.robustintelligence.com/blog-posts/failure-modes-when-productionizing-ai-systems
32. Notable runaway-optimiser-like LLM failure modes on Biologically ..., accessed on April 21, 2025, https://www.lesswrong.com/posts/PejNckwQj3A2MGhMA/notable-runaway-optimiser-like-llm-failure-modes-on
33. 18 Robust AI – Machine Learning Systems, accessed on April 21, 2025, https://mlsysbook.ai/contents/core/robust_ai/robust_ai.html
34. SegXAL: Explainable Active Learning for Semantic Segmentation in Driving Scene Scenarios - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2408.04482
35. The challenge of uncertainty quantification of large language models in medicine - arXiv, accessed on April 21, 2025, https://arxiv.org/pdf/2504.05278
36. Energy and Entropy based Intelligence Metric for Performance Estimation in DNNs | Request PDF - ResearchGate, accessed on April 21, 2025, https://www.researchgate.net/publication/369484680_Energy_and_Entropy_based_Intelligence_Metric_for_Performance_Estimation_in_DNNs
37. aif360.sklearn.metrics.generalized_entropy_error - Read the Docs, accessed on April 21, 2025, https://aif360.readthedocs.io/en/stable/modules/generated/aif360.sklearn.metrics.generalized_entropy_error.html
38. Probabilistic Modeling of Semantic Ambiguity for Scene Graph Generation - CVF Open Access, accessed on April 21, 2025, https://openaccess.thecvf.com/content/CVPR2021/papers/Yang_Probabilistic_Modeling_of_Semantic_Ambiguity_for_Scene_Graph_Generation_CVPR_2021_paper.pdf
39. (PDF) Epistemic Value Theory and Information Ethics - ResearchGate, accessed on April 21, 2025, https://www.researchgate.net/publication/226243255_Epistemic_Value_Theory_and_Information_Ethics
40. The Unreasonable Destructiveness of Political Correctness in Philosophy - MDPI, accessed on April 21, 2025, https://www.mdpi.com/2409-9287/2/3/17
41. Active Inference, Epistemic Value, and Uncertainty in Conceptual Disorganization in First-Episode Schizophrenia - PMC - PubMed Central, accessed on April 21, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC10031740
42. Epistemic Justification - Internet Encyclopedia of Philosophy, accessed on April 21, 2025, https://iep.utm.edu/epi-just/
43. Knowledge, Understanding and Epistemic Value - ResearchGate, accessed on April 21, 2025,

https://www.researchgate.net/publication/231992390_Knowledge_Understanding_and_Epistemic_Value
44. Epistemology in the Age of Large Language Models - MDPI, accessed on April 21, 2025, https://www.mdpi.com/2673-9585/5/1/3
45. Epistemic Alignment: A Mediating Framework for User-LLM Knowledge Delivery - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2504.01205v1
46. Epistemic AI, accessed on April 21, 2025, https://epistemic.ai/
47. Philosophy Eats AI - MIT Sloan Management Review, accessed on April 21, 2025, https://sloanreview.mit.edu/article/philosophy-eats-ai/
48. (PDF) Connecting ethics and epistemology of AI - ResearchGate, accessed on April 21, 2025, https://www.researchgate.net/publication/366165783_Connecting_ethics_and_epistemology_of_AI
49. Correctness, Artificial Intelligence, and the Epistemic Value of Mathematical Proof - PhilSci-Archive, accessed on April 21, 2025, https://philsci-archive.pitt.edu/24324/1/Epistemic_Value_of_Proof.pdf
50. Exploration via Epistemic Value Estimation | Proceedings of the AAAI Conference on Artificial Intelligence, accessed on April 21, 2025, https://ojs.aaai.org/index.php/AAAI/article/view/26164
51. Conflicts in Urban Future-Making - Governance, Institutions, and Transformative Change - SSOAR: Social Science Open Access Repository, accessed on April 21, 2025, https://www.ssoar.info/ssoar/bitstream/handle/document/98749/ssoar-2024-grubbauer_et_al-Conflicts_in_Urban_Future-Making_Governance.pdf?sequence=1&isAllowed=y&lnkname=ssoar-2024-grubbauer_et_al-Conflicts_in_Urban_Future-Making_Governance.pdf
52. (PDF) Who's Afraid of Adversariality? Conflict and Cooperation in Argumentation, accessed on April 21, 2025, https://www.researchgate.net/publication/347813444_Who's_Afraid_of_Adversariality_Conflict_and_Cooperation_in_Argumentation
53. Conflict Data Fusion in a Multi-Agent System Premised on the Base Basic Probability Assignment and Evidence Distance - PubMed Central, accessed on April 21, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC8308205/
54. Who's Afraid of Adversariality? Conflict and Cooperation in Argumentation - PMC, accessed on April 21, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC8556168/
55. Quantifying Misalignment Between Agents: Towards a ..., accessed on April 21, 2025, https://ojs.aaai.org/index.php/AAAI/article/view/34947
56. AI for Explaining Decisions in Multi-Agent Environments | Request PDF - ResearchGate, accessed on April 21, 2025, https://www.researchgate.net/publication/342544354_AI_for_Explaining_Decisions_in_Multi-Agent_Environments
57. Quantifying Misalignment Between Agents: Towards a Sociotechnical Understanding of Alignment - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2406.04231v3

58. Multi-Agent Systems & Collaborative AI: The Power of Working Together, accessed on April 21, 2025, https://jaimlr.github.io/Journal-of-Artificial-Intelligence-Machine-Learning-Research/multi-agent-systems-and-collaborative-ai-by-loveleen-narang.html
59. What Is Multi-Agent AI? Definition, Benefits, and Examples - New Horizons - Blog, accessed on April 21, 2025, https://www.newhorizons.com/resources/blog/multi-agent-ai
60. Full article: Deciding on the Tit for the Tat: Decision-Making in the Wake of Ceasefire Violations - Taylor & Francis Online, accessed on April 21, 2025, https://www.tandfonline.com/doi/full/10.1080/13533312.2021.1914596
61. Multiagent Systems: A Survey from a Machine Learning Perspective - CMU School of Computer Science, accessed on April 21, 2025, https://www.cs.cmu.edu/~mmv/papers/MASsurvey.pdf
62. isaiahbjork/chain-of-thought · Datasets at Hugging Face, accessed on April 21, 2025, https://huggingface.co/datasets/isaiahbjork/chain-of-thought/viewer
63. Artificial Intelligence for Automated Decision-Making in Error-Pattern Recognition, accessed on April 21, 2025, https://edoc.ub.uni-muenchen.de/34029/1/Guimera_Cuevas_felip.pdf
64. Task cognition and planning for service robots - OAE Publishing Inc., accessed on April 21, 2025, https://www.oaepublish.com/articles/ir.2025.08
65. Agent-driven Generative Semantic Communication with Cross-Modality and Prediction | Bytez, accessed on April 21, 2025, https://bytez.com/docs/arxiv/2404.06997/paper
66. Towards a Framework for Semantic Information, accessed on April 21, 2025, https://theinformationalturn.net/wp-content/uploads/2012/12/towards_a_framework_for_semantic_information.pdf
67. Deep Learning-Enabled Semantic Communication Systems with Task-Unaware Transmitter and Dynamic Data - ResearchGate, accessed on April 21, 2025, https://www.researchgate.net/publication/360333590_Deep_Learning-Enabled_Semantic_Communication_Systems_with_Task-Unaware_Transmitter_and_Dynamic_Data
68. arXiv:2405.01379v4 [cs.CL] 11 Oct 2024, accessed on April 21, 2025, https://arxiv.org/pdf/2405.01379
69. End to End Learning in Autonomous Driving Systems - UC Berkeley EECS, accessed on April 21, 2025, https://www2.eecs.berkeley.edu/Pubs/TechRpts/2020/EECS-2020-5.pdf
70. Verification and Refinement of Natural Language Explanations through LLM-Symbolic Theorem Proving - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2405.01379v3
71. Research on the Application of Artificial Intelligence-Driven Cross-Modal Semantic Communication System in the Tourism Industry - Atlantis Press, accessed on April 21, 2025, https://www.atlantis-press.com/article/126007725.pdf
72. (PDF) Towards Semantic Communications: Deep Learning-Based Image Semantic Coding - ResearchGate, accessed on April 21, 2025, https://www.researchgate.net/publication/362567393_Towards_Semantic_Commu

nications_Deep_Learning-Based_Image_Semantic_Coding
73. Ernie-Gram BiGRU Attention: An Improved Multi-Intention Recognition Model for Air Traffic Control - MDPI, accessed on April 21, 2025, https://www.mdpi.com/2226-4310/10/4/349
74. Deep Dive on Sankey Diagrams - Plotly, accessed on April 21, 2025, https://plotly.com/blog/sankey-diagrams/
75. Sankey diagram - From data to Viz, accessed on April 21, 2025, https://www.data-to-viz.com/graph/sankey.html
76. Data Flow Visualization Through Sankey Diagram. - Think Design, accessed on April 21, 2025, https://think.design/services/data-visualization-data-design/sankey-diagram/
77. what is a sankey diagram? - storytelling with data, accessed on April 21, 2025, https://www.storytellingwithdata.com/blog/what-is-a-sankey-diagram
78. Overview of Sankey Flow Diagrams: Focusing on Symptom Trajectories in Older Adults with Advanced Cancer, accessed on April 21, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC9232856/
79. Visualize data density and histograms with heat maps - Chronosphere Docs, accessed on April 21, 2025, https://docs.chronosphere.io/observe/dashboards/panels/heat-map
80. (PDF) Visualizing Uncertainty in Node-Link Diagrams - a User Study - ResearchGate, accessed on April 21, 2025, https://www.researchgate.net/publication/318157533_Visualizing_Uncertainty_in_Node-Link_Diagrams_-_a_User_Study
81. Visualizing node attribute uncertainty in graphs, accessed on April 21, 2025, https://avis.soe.ucsc.edu/bullseye.html
82. 16 Visualizing uncertainty - Fundamentals of Data Visualization, accessed on April 21, 2025, https://clauswilke.com/dataviz/visualizing-uncertainty.html
83. Visualizing Uncertainty - GitHub Pages, accessed on April 21, 2025, https://friendly.github.io/6135/lectures/Uncertainty.pdf
84. Visualizing Uncertainty in Sets - A Rostock, accessed on April 21, 2025, https://vca.informatik.uni-rostock.de/~ct/publications/Tominski23VisUncertaintySets.pdf
85. Set Visualization and Uncertainty - DROPS - Schloss Dagstuhl, accessed on April 21, 2025, https://drops.dagstuhl.de/storage/04dagstuhl-reports/volume12/issue11/22462/DagRep.12.11.66/DagRep.12.11.66.pdf
86. Uncertainty Visualization: Concepts, Methods, and Applications in Biological Data Visualization - PMC - PubMed Central, accessed on April 21, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC9580861/
87. Visualizing uncertainty in flow diagrams: a case study in product costing - ResearchGate, accessed on April 21, 2025, https://www.researchgate.net/publication/319133132_Visualizing_uncertainty_in_flow_diagrams_a_case_study_in_product_costing
88. VISIT: Visualizing and Interpreting the Semantic Information Flow of Transformers - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2305.13417

89. An empirical analysis of error propagation in critical software systems - ResearchGate, accessed on April 21, 2025, https://www.researchgate.net/publication/339924246_An_empirical_analysis_of_error_propagation_in_critical_software_systems
90. Modeling and measurement of error propagation in a multimodule computing system - RTCL - University of Michigan, accessed on April 21, 2025, https://rtcl.eecs.umich.edu/rtclweb/assets/publications/1988/shin1988modeling.pdf
91. Error propagation: Tracing the Ripple Effects of Errors in Complex Systems update, accessed on April 21, 2025, https://fastercapital.com/content/Error-propagation--Tracing-the-Ripple-Effects-of-Errors-in-Complex-Systems-update.html
92. Error Propagation Analysis for File Systems, accessed on April 21, 2025, https://web.cs.ucdavis.edu/~rubio/includes/pldi09.pdf
93. Advances and challenges in semantic communications: A systematic review | National Science Open (NSO), accessed on April 21, 2025, https://www.nso-journal.org/articles/nso/full_html/2024/04/NSO20230029/NSO20230029.html
94. Bridging the Gap Between Modular and End-to-end Autonomous Driving Systems - UC Berkeley EECS, accessed on April 21, 2025, https://www2.eecs.berkeley.edu/Pubs/TechRpts/2022/EECS-2022-79.pdf
95. Understanding error propagation in deep learning neural network (DNN) accelerators and applications | Request PDF - ResearchGate, accessed on April 21, 2025, https://www.researchgate.net/publication/320955461_Understanding_error_propagation_in_deep_learning_neural_network_DNN_accelerators_and_applications
96. Modular Fault Diagnosis Framework for Complex Autonomous Driving Systems - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2411.09643v1
97. Neuro-Symbolic Artificial Intelligence (AI) for Intent based Semantic Communication - arXiv, accessed on April 21, 2025, https://arxiv.org/pdf/2205.10768
98. Cocoa: Co-Planning and Co-Execution with AI Agents - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2412.10999v3
99. Prompt Recursive Search: A Living Framework with Adaptive Growth in LLM Auto-Prompting - arXiv, accessed on April 21, 2025, https://arxiv.org/pdf/2408.01423
100. [2408.01423] Prompt Recursive Search: A Living Framework with Adaptive Growth in LLM Auto-Prompting - arXiv, accessed on April 21, 2025, https://arxiv.org/abs/2408.01423
101. Emergent Abilities in Large Language Models: A Survey - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2503.05788v2
102. Gödel Agent: A Self-Referential Framework for Agents Recursively Self-Improvement - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2410.04444v2
103. Large Language Model Agent: A Survey on Methodology, Applications and

Challenges, accessed on April 21, 2025, https://arxiv.org/html/2503.21460
104. Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers) - ACL Anthology, accessed on April 21, 2025, https://aclanthology.org/volumes/2024.naacl-long/
105. Scaling Test-Time Reasoning in Large Language Models Through Logic Unit Alignment - arXiv, accessed on April 21, 2025, https://www.arxiv.org/pdf/2502.07803
106. NeurIPS Poster Buffer of Thoughts: Thought-Augmented Reasoning with Large Language Models, accessed on April 21, 2025, https://neurips.cc/virtual/2024/poster/96264
107. [2502.14143] Multi-Agent Risks from Advanced AI - arXiv, accessed on April 21, 2025, https://arxiv.org/abs/2502.14143
108. Buffer of Thoughts: Thought-Augmented Reasoning with Large Language Models - NIPS papers, accessed on April 21, 2025, https://proceedings.neurips.cc/paper_files/paper/2024/file/cde328b7bf6358f5ebb91fe9c539745e-Paper-Conference.pdf
109. Language Model Agents Suffer from Compositional Generalization in Web Automation - OpenReview, accessed on April 21, 2025, https://openreview.net/pdf?id=acgxF5AVLq
110. Hallucination Mitigation for Retrieval-Augmented Large Language Models: A Review - MDPI, accessed on April 21, 2025, https://www.mdpi.com/2227-7390/13/5/856
111. The AI-Based Cyber Threat Landscape: A Survey - ResearchGate, accessed on April 21, 2025, https://www.researchgate.net/publication/339081899_The_AI-Based_Cyber_Threat_Landscape_A_Survey
112. Cascading Failure Analysis on Shanghai Metro Networks: An Improved Coupled Map Lattices Model Based on Graph Attention Networks - MDPI, accessed on April 21, 2025, https://www.mdpi.com/1660-4601/19/1/204
113. A Methodology for Evaluating Operator Usage of Machine Learning Recommendations for Power Grid Contingency Analysis, accessed on April 21, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC9237339/
114. A Proposal for Infrastructure Adaptation and Cascading Failures for Black Swans Ryan Michael Hoff A Dissertation Presented in Pa - cloudfront.net, accessed on April 21, 2025, https://d1rbsgppyrdqq4.cloudfront.net/s3fs-public/c7/Hoff_asu_0010E_23035.pdf?versionId=U_2MLbqc_EFlp39cIVMZZS3zRUI1jQBR&X-Amz-Content-Sha256=UNSIGNED-PAYLOAD&X-Amz-Algorithm=AWS4-HMAC-SHA256&X-Amz-Credential=AKIASBVQ3ZQ4YNQVYJLW/20250321/us-west-2/s3/aws4_request&X-Amz-Date=20250321T175537Z&X-Amz-SignedHeaders=host&X-Amz-Expires=120&X-Amz-Signature=c02b74b36805a6b8df5210b1ccb684c38353fe6c7bb75c654be5d3904c45ede1
115. Potential smart grid vulnerabilities to cyber attacks: Current threats and existing mitigation strategies - PMC - PubMed Central, accessed on April 21,

2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC11470553/
116. Full article: Application of clustering algorithms for dimensionality reduction in infrastructure resilience prediction models - Taylor & Francis Online, accessed on April 21, 2025, https://www.tandfonline.com/doi/full/10.1080/15732479.2024.2366958
117. Full article: Grand challenges in industrial and systems engineering - Taylor & Francis Online, accessed on April 21, 2025, https://www.tandfonline.com/doi/full/10.1080/00207543.2024.2432463
118. G.Cigdem Yalcin - Google Scholar, accessed on April 21, 2025, https://scholar.google.com/citations?user=uHujbeUAAAAJ&hl=it
119. Artificial intelligence techniques for stability analysis in modern power systems - SciOpen, accessed on April 21, 2025, https://www.sciopen.com/article/10.23919/IEN.2024.0027
120. Multi-Agent Systems and Complex Networks: Review and Applications in Systems Engineering - MDPI, accessed on April 21, 2025, https://www.mdpi.com/2227-9717/8/3/312
121. Survey on Simulation and Vulnerability Testing in Smart Grid - Preprints.org, accessed on April 21, 2025, https://www.preprints.org/manuscript/202402.0047
122. A Survey on Vulnerability Prioritization: Taxonomy, Metrics, and Research Challenges, accessed on April 21, 2025, https://arxiv.org/html/2502.11070v1
123. Papers by Dilek Küçük - AIModels.fyi, accessed on April 21, 2025, https://www.aimodels.fyi/authors/arxiv/Dilek%20K%C3%BC%C3%A7%C3%BCk
124. Jinzhu Yu - Google Scholar, accessed on April 21, 2025, https://scholar.google.com/citations?user=UP-znn8AAAAJ&hl=en
125. Jinzhu Yu - Google Scholar, accessed on April 21, 2025, https://scholar.google.com.vn/citations?user=UP-znn8AAAAJ&hl=vi
126. 10 Visualization Techniques for Complex Spatial Relationships That Reveal Hidden Patterns - Map Library - Lovin' Cartography, accessed on April 21, 2025, https://www.maplibrary.org/1459/visualization-techniques-for-complex-spatial-relationships/
127. Build multi-agent systems with LangGraph and Amazon Bedrock - AWS, accessed on April 21, 2025, https://aws.amazon.com/blogs/machine-learning/build-multi-agent-systems-with-langgraph-and-amazon-bedrock/
128. Full article: An automated construction method of 3D knowledge graph based on multi-agent systems in virtual geographic scene - Taylor & Francis Online, accessed on April 21, 2025, https://www.tandfonline.com/doi/full/10.1080/17538947.2024.2449185
129. Using multi-agent system for code and data propagation - DiVA portal, accessed on April 21, 2025, https://www.diva-portal.org/smash/get/diva2:831748/FULLTEXT01.pdf
130. Node-Link Graph Widgets, accessed on April 21, 2025, https://docs.logrhythm.com/lrsiem/docs/node-link-graph-widgets
131. Statistical Data Visualization: Node - Link Diagrams - GitHub Pages, accessed on April 21, 2025, https://krisrs1128.github.io/stat479/posts/2021-03-06-week8-2/

132. jhoffswell.github.io, accessed on April 21, 2025, https://jhoffswell.github.io/website/resources/papers/2022-ViSRE-VISSOFT.pdf
133. 17 Important Data Visualization Techniques | HBS Online, accessed on April 21, 2025, https://online.hbs.edu/blog/post/data-visualization-techniques
134. A Complete Guide to Heatmaps | Atlassian, accessed on April 21, 2025, https://www.atlassian.com/data/charts/heatmap-complete-guide
135. Heat map visualization in a dashboard - IBM, accessed on April 21, 2025, https://www.ibm.com/docs/en/cognos-analytics/11.2.0?topic=types-heat-map
136. Heatmap Widget - Broadcom TechDocs, accessed on April 21, 2025, https://techdocs.broadcom.com/us/en/vmware-cis/aria/aria-operations/8-16/vmware-aria-operations-configuration-guide-8-16/using-widgets/widget-definitions-list/heat-map-widget.html
137. Heatmap visualization example - ServiceNow, accessed on April 21, 2025, https://www.servicenow.com/docs/bundle/yokohama-now-intelligence/page/use/par-for-workspace/task/dv-example-heatmap.html
138. Quantitative Methods for Uncertainty Visualization - KOPS, accessed on April 21, 2025, https://kops.uni-konstanz.de/bitstreams/de925c8a-2b86-45ba-91b6-0d918f621690/download
139. BERTScore in AI: Transforming Semantic Text Evaluation and Quality - Galileo AI, accessed on April 21, 2025, https://www.galileo.ai/blog/bert-score-explained-guide
140. Different Techniques for Sentence Semantic Similarity in NLP | GeeksforGeeks, accessed on April 21, 2025, https://www.geeksforgeeks.org/different-techniques-for-sentence-semantic-similarity-in-nlp/
141. Semantic Textual Similarity — Sentence Transformers documentation, accessed on April 21, 2025, https://sbert.net/docs/sentence_transformer/usage/semantic_textual_similarity.html
142. Semantic similarity prediction is better than other semantic similarity measures - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2309.12697v2
143. Semantic Density: Uncertainty Quantification for Large Language Models through Confidence Measurement in Semantic Space - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2405.13845v3
144. Semantic Drift in Multilingual Representations | Computational Linguistics - MIT Press Direct, accessed on April 21, 2025, https://direct.mit.edu/coli/article/46/3/571/93376/Semantic-Drift-in-Multilingual-Representations
145. Characterizing and Measuring Linguistic Dataset Drift - ACL Anthology, accessed on April 21, 2025, https://aclanthology.org/2023.acl-long.498.pdf
146. Understanding Machine Learning Accuracy: Metrics and Methods to Measure Model Performance | Udacity, accessed on April 21, 2025, https://www.udacity.com/blog/2024/12/understanding-machine-learning-accuracy-metrics-and-methods-to-measure-model-performance.html

147. Metrices in Machine learning - Corpnce, accessed on April 21, 2025, https://www.corpnce.com/metrices-in-machine-learning/
148. 5 Types of Data Visualization (With Examples) - Built In, accessed on April 21, 2025, https://builtin.com/articles/types-of-data-visualization
149. Interpretable Self-Aware Neural Networks for Robust Trajectory Prediction - Proceedings of Machine Learning Research, accessed on April 21, 2025, https://proceedings.mlr.press/v205/itkina23a/itkina23a.pdf
150. (PDF) Gradient-Based Quantification of Epistemic Uncertainty for Deep Object Detectors, accessed on April 21, 2025, https://www.researchgate.net/publication/353170983_Gradient-Based_Quantification_of_Epistemic_Uncertainty_for_Deep_Object_Detectors
151. Uncertainty-Aware Multimodal Trajectory Prediction via a Single Inference from a Single Model - PMC, accessed on April 21, 2025, https://pmc.ncbi.nlm.nih.gov/articles/PMC11723313/
152. Second-Order Uncertainty Quantification: A Distance-Based Approach - GitHub, accessed on April 21, 2025, https://raw.githubusercontent.com/mlresearch/v235/main/assets/sale24a/sale24a.pdf
153. NeurIPS Poster Are Uncertainty Quantification Capabilities of Evidential Deep Learning a Mirage?, accessed on April 21, 2025, https://neurips.cc/virtual/2024/poster/95329
154. Are Uncertainty Quantification Capabilities of Evidential Deep Learning a Mirage? - NIPS papers, accessed on April 21, 2025, https://proceedings.neurips.cc/paper_files/paper/2024/file/c3177be226ee12e34d6ba3b5e6fe6a5b-Paper-Conference.pdf
155. Delta Variances - Google DeepMind, accessed on April 21, 2025, https://deepmind.google/research/publications/112791/
156. 1 INTRODUCTION - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2312.00995v1
157. A Survey on Uncertainty Quantification of Large Language Models: Taxonomy, Open Research Challenges, and Future Directions - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2412.05563v1
158. Achieving well-informed decision-making in drug discovery: a comprehensive calibration study using neural network-based - Chalmers Research, accessed on April 21, 2025, https://research.chalmers.se/publication/545577/file/545577_Fulltext.pdf
159. Agents for good? Reconciling agentic AI with existing AI governance frameworks - techUK, accessed on April 21, 2025, https://www.techuk.org/resource/agents-for-good-reconciling-agentic-ai-with-existing-ai-governance-frameworks.html
160. The evolving ethics and governance landscape of agentic AI | IBM, accessed on April 21, 2025, https://www.ibm.com/think/insights/ethics-governance-agentic-ai
161. Responsible adoption of AI agents in the enterprise: PwC, accessed on April 21, 2025,

https://www.pwc.com/us/en/tech-effect/ai-analytics/responsible-ai-agents.html
162. IBM's answer to governing AI Agents: Automation and Evaluation with watsonx.governance, accessed on April 21, 2025, https://www.ibm.com/new/announcements/ibms-answer-to-governing-ai-agents-automation-and-evaluation-with-watsonx-governance
163. Learn 25 Dashboard Design Principles & BI Best Practices - RIB Software, accessed on April 21, 2025, https://www.rib-software.com/en/blogs/bi-dashboard-design-principles-best-practices
164. 4 Dashboard Design Principles for Better Data Visualization - Sisense, accessed on April 21, 2025, https://www.sisense.com/blog/4-design-principles-creating-better-dashboards/
165. How to Build THE Perfect Data Analytics Dashboard - Triple Whale, accessed on April 21, 2025, https://www.triplewhale.com/blog/build-the-perfect-data-analytics-dashboard
166. Real-time Data Visualization: How to build faster dashboards - Tinybird, accessed on April 21, 2025, https://www.tinybird.co/blog-posts/real-time-data-visualization
167. Dashboard Interface Design Agency | Fuselab Creative, accessed on April 21, 2025, https://fuselabcreative.com/intelligent-interface/
168. Dashboard Tools for Data Analytics - DATAVERSITY, accessed on April 21, 2025, https://www.dataversity.net/dashboard-tools-for-data-analytics/
169. AgentLens: Visual Analysis for Agent Behaviors in LLM-based Autonomous Systems - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2402.08995v1
170. MARLens: Understanding Multi-agent Reinforcement Learning for Traffic Signal Control via Visual Analytics - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2410.01364v1
171. Cooperative and Competitive Multi-Agent Systems: From Optimization to Games, accessed on April 21, 2025, https://www.ieee-jas.net/article/doi/10.1109/JAS.2022.105506
172. Visualize Your Analytics with Interactive Dashboards | Oracle Jordan, accessed on April 21, 2025, https://www.oracle.com/jo/business-analytics/analytics-platform/capabilities/visualize/
173. Visual Analytics: How Do You Bring Data to Life? - ChartExpo, accessed on April 21, 2025, https://chartexpo.com/blog/visual-analytics
174. Visualizing data effectively on dashboards - Esri, accessed on April 21, 2025, https://www.esri.com/arcgis-blog/products/ops-dashboard/decision-support/visualizing-data-effectively-on-dashboards/
175. Nine Best Practices for Big Data Dashboards Using SAS® Visual Analytics, accessed on April 21, 2025, https://support.sas.com/resources/papers/proceedings17/SAS0517-2017.pdf
176. Dashboard Chart Actions from multiple data layers - Esri Community, accessed on April 21, 2025, https://community.esri.com/t5/arcgis-dashboards-questions/dashboard-chart-ac

tions-from-multiple-data-layers/td-p/1478675
177. AgentLens: Visual Analysis for Agent Behaviors in LLM-based Autonomous Systems - arXiv, accessed on April 21, 2025, https://arxiv.org/abs/2402.08995
178. [论文审查] AgentLens: Visual Analysis for Agent Behaviors in LLM-based Autonomous Systems - Moonlight, accessed on April 21, 2025, https://www.themoonlight.io/zh/review/agentlens-visual-analysis-for-agent-behaviors-in-llm-based-autonomous-systems
179. AgentLens: Visual Analysis for Agent Behaviors in LLM-based Autonomous Systems | Request PDF - ResearchGate, accessed on April 21, 2025, https://www.researchgate.net/publication/380323142_AgentLens_Visual_Analysis_for_Agent_Behaviors_in_LLM-based_Autonomous_Systems
180. AgentCoord: Visually Exploring Coordination Strategy for LLM-based Multi-Agent Collaboration - ResearchGate, accessed on April 21, 2025, https://www.researchgate.net/publication/380895414_AgentCoord_Visually_Exploring_Coordination_Strategy_for_LLM-based_Multi-Agent_Collaboration
181. Bo Pan - Google Scholar, accessed on April 21, 2025, https://scholar.google.com/citations?user=eE2yNGUAAAAJ&hl=en
182. kyegomez/awesome-multi-agent-papers - GitHub, accessed on April 21, 2025, https://github.com/kyegomez/awesome-multi-agent-papers
183. AgentCoord: Visually Exploring Coordination Strategy for LLM-based Multi-Agent Collaboration - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2404.11943v1
184. Exploring Autonomous Agents through the Lens of Large Language Models: A Review, accessed on April 21, 2025, https://arxiv.org/html/2404.04442v1
185. Towards Effective GenAI Multi-Agent Collaboration: Design and Evaluation for Enterprise Applications - Amazon Science, accessed on April 21, 2025, https://assets.amazon.science/bc/1e/1202475d44a6842a065dd4adf9b9/towards-effective-genai-multi-agent-collaboration-design-and-evaluation-for-enterprise-applications.pdf
186. [Literature Review] AgentLens: Visual Analysis for Agent Behaviors, accessed on April 21, 2025, https://www.themoonlight.io/en/review/agentlens-visual-analysis-for-agent-behaviors-in-llm-based-autonomous-systems
187. Literature Review — AutoGen - Microsoft Open Source, accessed on April 21, 2025, https://microsoft.github.io/autogen/dev/user-guide/agentchat-user-guide/examples/literature-review.html
188. Epidemic Modeling with Generative Agents (2023) | Ross Williams, accessed on April 21, 2025, https://typeset.io/papers/epidemic-modeling-with-generative-agents-38sxij9h
189. VISIT: Visualizing and Interpreting the Semantic Information Flow of Transformers - Yonatan Belinkov, accessed on April 21, 2025, https://belinkov.com/assets/pdf/emnlp2023-findings-visit.pdf
190. VISIT: Visualizing and Interpreting the Semantic Information Flow of Transformers, accessed on April 21, 2025,

https://openreview.net/forum?id=7O9bTjLgTQ¬eId=siAuLN297A
191. [PDF] VISIT: Visualizing and Interpreting the Semantic Information Flow of Transformers, accessed on April 21, 2025, https://www.semanticscholar.org/paper/5d9401adc6ce7e0d538ddbb44253ebe413795c62
192. Generating and Visualizing Context Vectors in Transformers - MachineLearningMastery.com, accessed on April 21, 2025, https://machinelearningmastery.com/generating-and-visualizing-context-vectors-in-transformers/
193. VISIT: Visualizing and Interpreting the Semantic Information Flow of Transformers - arXiv, accessed on April 21, 2025, https://arxiv.org/abs/2305.13417
194. shacharKZ/VISIT-Visualizing-Transformers - GitHub, accessed on April 21, 2025, https://github.com/shacharKZ/VISIT-Visualizing-Transformers
195. Transformers Meet Visual Learning Understanding: A Comprehensive Review - arXiv, accessed on April 21, 2025, https://arxiv.org/pdf/2203.12944
196. Explaining Information Flow Inside Vision Transformers Using Markov Chain, accessed on April 21, 2025, https://xai4debugging.github.io/files/papers/explaining_information_flow_in.pdf
197. Transformer Interpretability Beyond Attention Visualization - CVF Open Access, accessed on April 21, 2025, https://openaccess.thecvf.com/content/CVPR2021/papers/Chefer_Transformer_Interpretability_Beyond_Attention_Visualization_CVPR_2021_paper.pdf
198. arXiv:2502.12033v1 [cs.LG] 17 Feb 2025, accessed on April 21, 2025, https://arxiv.org/pdf/2502.12033
199. Provenance-Enabled Explainable AI - Benjamin E. Ujcich, accessed on April 21, 2025, https://personal.benujcich.georgetown.domains/papers/25_SIGMOD.pdf
200. Enhancing Intent Understanding for Ambiguous Prompts through Human-Machine Co-Adaptation - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2501.15167v1
201. Exploring Prompt Engineering: A Systematic Review with SWOT Analysis - arXiv, accessed on April 21, 2025, https://arxiv.org/html/2410.12843v1
202. OVERCOMING THE AMBIGUITY REQUIREMENT USING GENERATIVE AI, accessed on April 21, 2025, https://mdh.diva-portal.org/smash/get/diva2:1931301/FULLTEXT01.pdf
203. Prompt Engineering: 8 Advanced Techniques for Better AI Results - Nanonets, accessed on April 21, 2025, https://nanonets.com/blog/prompt-engineering/
204. Automatic Prompt Engineering: Boosting AI Performance with Smarter Prompts - Portkey, accessed on April 21, 2025, https://portkey.ai/blog/what-is-automated-prompt-engineering